Exercise 4:

1 SQL Joins

1. SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees e
INNER JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

2. SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees e
LEFT JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

3. SELECT p.ProjectID, p.ProjectName, p.Budget, p.Status,
e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary
FROM Employees e
RIGHT JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

4. SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees e
FULL OUTER JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

2. 5. SELECT City AS Location FROM Employees
UNION
SELECT Status AS Location FROM Projects;

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6. SELECT City AS Location FROM Employees
UNION ALL
SELECT Status AS Location FROM Projects;

3. FILTERING

7. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70000;

8. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');

9. SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Status <> 'Completed';

10. SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000
AND Status <> 'Completed';

11. SELECT EmployeeID, FirstName, LastName, Department, Salary, City
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;

12. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;

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13.
```
SELECT Department, SUM(Salary) AS totalSalary
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;
```

14.
```
SELECT City, AVG(Salary) AS AverageSalary
FROM Employee
GROUP BY City
HAVING AVG(Salary) > 60000;
```

15.
```
SELECT Department, COUNT(EmployeeID) AS Employee Count
FROM Employees
GROUP BY Department
HAVING COUNT(EmployeeID) >1;
```

16.
```
SELECT Status, COUNT(projectID) AS ProjectCount
FROM Projects
GROUP BY Status
HAVING COUNT(ProjectID)>=2;
```

17.
```
SELECT e.EmployeeID, e.FirstName, e.LastName,
       SUM(p.Budget) AS Total Project Budget
FROM Employee e
JOIN Projects p
  ON e.EmployeeID = p.EmployeeID
GROUP BY e.EmployeeID, e.FirstName, e.LastName
HAVING SUM(p.Budget) > 150000;
```

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17.

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	150000

16.

Status	Project Count
Completed	2
Ongoing	2
Pending	2

15.

Department	Employee Count
IT	2
Finance	2

14.

City	Average Salary
New York	75000
Toronto	75000
London	72000

13.

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

12.

EmployeeID	FirstName	LastName	Dept	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	Dana	Brown	IT	72000

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11.

EmployeeID	FirstName	LastName	Dept	Salary	City
6	Michael	Clarice	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	7000	New York

10.

ProjectID	ProjectName	Budget	Status
103	CybersecurityAudit	75000	Pending
104	Financial Analysis	9000	Ongoing
105	Risk Management	8000	Pending

9.

ProjectID	Project Name	Budget	Status
102	Employee Training	5000	Ongoing
103	Cybersec Audit	7500	Pending
104	Financial Analysis	9000	Ongoing
106	Risk Management	8000	Pending

8.

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	7000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

7.

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

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